Notebook 1 = Establishing the Data

- ***Course:*** DS 2023 – Communicating with Data
- ***Project Title:*** How Audio Features(Danceability) Shape Genre Popularity on Spotify
- ***Date:*** December 12, 2025

This notebook's function is to establish the structure, source, and general info relating to the Spotify audio features dataset used throughout this project.

In [1]:
```
#1 Acquire and read in the data source(s).

#imports
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

#my spotify features dataset
df = pd.read_csv("SpotifyFeatures.csv")

#dataset basic structure
df.head()
df = df.fillna(0)
```

2. Describe How to Get the Data

To obtain the "SpotifyFeatures" dataset:

Dataset Link:
https://www.kaggle.com/datasets/zaheenhamidani/ultimate-spotify-tracks

1. Open the Kaggle link above.
2. Log in/ Sign up for a Kaggle account (this is required to download files).
3. Click the **Download** button on the top right of the dataset page.
4. Save the ZIP file to your computer.
5. Upload the CSV file into your Jupyter notebook.
6. Proceed to load the data using: df = pd.read_csv("SpotifyFeatures.csv")

3. Describe who produced the data and how

This project uses the *Ultimate Spotify Tracks Dataset* which was created and published by Kaggle user **'zaheenhamidani'** According to the dataset's Kaggle description, the data was collected using Spotify's Web API and was last updated 6 years ago in October,2019. Each observation within this dataset represents an individual song with associated musical attributes measured by Spotify.

4. Describe the Data’s Features with a COLS Table

Column Name	Description	Type
artist_name	Name of the artist who performed the track	object
track_name	Title of the track	object
track_id	Unique Spotify ID assigned to each track	object
popularity	Spotify popularity score (0–100)	int
acousticness	Measure of how acoustic a track is (0.0–1.0)	float
danceability	How suitable a track is for dancing (0.0–1.0)	float
duration_ms	Track duration in milliseconds	int
energy	Measure of intensity/energy in the track (0.0–1.0)	float
instrumentalness	Likelihood the track contains no vocals (0.0–1.0)	float
key	Musical key the track is in (0–11)	int
liveness	Likelihood that the track was performed live (0.0–1.0)	float
loudness	Track loudness in decibels (dB)	float
mode	Musical mode: 1 = major, 0 = minor	int
speechiness	Presence of spoken words in the track (0.0–1.0)	float
tempo	Estimated tempo in beats per minute (BPM)	float
valence	Musical positivity/happiness (0.0–1.0)	float
genre	Assigned genre tag for the track	object
release_year	Year the track was released	int